



MODULE/COURSE OUTLINE

Faculty	Science	
Department	Mathematics	
Programme	Bachelor of Science in Mathematics	
Course/Module Title and Code	MAT111 INTRODUCTORY MATHEMATICS I	
Credits	12	
NCQF Level	5	
Type	Core	
Semester	1	
Prerequisites	None	
Course Synopsis	This course is a pre-calculus course that consolidates and builds upon school mathematics. It will lead the student to a better understanding of algebraic concepts and techniques. It covers Functions, Polynomials, Exponential and logarithmic functions, Trigonometry and Complex numbers.	
Related Exit Level Outcome(s)	LO1. Demonstrate wider knowledge and problem-solving skills in the area of Mathematics	
Related Assessment Criteria	AC1.1 Demonstrate a general understanding of the basic principles of mathematics. AC1.2 Demonstrate knowledge through analysis of different forms of mathematical data.	
Course Learning Outcomes	CLO1 Demonstrate competence with different types and properties of functions, their inverses and graphs. CLO2 Understand basic concepts of trigonometry CLO3 Perform arithmetic operations on complex numbers	
Lecturer	Prof. Muatjetjeja, Dr R Tshelametse, B. Lephodisa, Dr O Daman, Dr TG Motsumi, Dr M. Sangago, M. Moetele, Dr. M. Koorapetse, Dr. C. Mabenga, Dr. S. Basimanebotlhe, Dr O. Dira, Dr E.R. Offen, Mr P. Moile, Mr O. Ramatebele	
Course Learning Outcome	Specific Learning Outcome	Topics and sub topics
CLO1. Demonstrate competence with different types and properties of functions, their inverses and graphs.	1.1 Determine range and domain of functions 1.2 Identify one-to-one functions 1.3 Find piecewise domains 1.4 Sketch graphs of functions 1.5 Find inverse of a function and explain the graphical relationship between the function and its inverse 1.6 Identify quadratic functions and polynomials 1.7 Solve equations involving exponential and logarithmic functions	• Functions - Determining range, domain, one-to-one, onto, equality of functions - Algebra of functions and composition - Piecewise domains - Graphs of functions - Inverse functions (including graphical relationship between a function and its inverse)

	1.8 Apply Remainder theorem and Factor theorem to find factors of a polynomial 1.9 Perform translations and reflections of graphs of a function.	- Quadratic functions - Polynomials - introduction, long division, algebra and graphs of rational functions - Exponential and logarithmic functions - Remainder theorem and Factor theorem - Translations and reflections (parallel to axes) of graphs of functions.
CLO2. Understand Basic Concepts of Trigonometry	2.1 Derive basic trigonometric identities 2.2 Derive cosine and sine rules 2.3 Memorize sum rules and double angle formulas 2.4 Determine maximum and minimum of trig expressions 2.5 Solve simple trigonometric equations 2.6 Find inverse of a trig function	• Trigonometry - Important angles, sin/cos/tan and their basic identities - Sine rule, cos rule, sums and differences of angles, double angle formula, half angle formula - The expression $\arccos(k + l\theta)$ and max/min of functions using expressions like this - Solution of simple trigonometric equations, cot/sec/cosec - Sin/cos/tan functions - periods, zeros, maxima and minima, inverse trig functions.
CLO3. Perform arithmetic operations on complex numbers	3.1 Define complex number and explain Argand diagrams 3.2 Apply de Moivre's theorem to trigonometric identities 3.3 Find roots of a complex number.	• Complex numbers - Introduction to complex numbers - Argand diagrams - De's theorem (including applications to trigonometric identities) and roots of a complex number.
Teaching and Learning strategies		Assessment strategies and activities
A combination of lectures and problem-solving type activities will be used. Lectures will be supported by power point presentation and other technology aided means.		Quizzes,
A combination of lectures and guided group discussion methods		Tests(at least two)
Tutorials: Tutorials will be used to strengthen and to monitor progress of learning. For this purpose a large lecture group will be divided into small groups		Exam: The exam covers all the contents (Summative Assessment)
Required Resources	Prescribed Text books	
	Introductory Mathematics, Customized Version for Univ. Botswana (2014), Cengage Learning ISBN 978-1-4737-5405-8	
	Recommended Reading List	
	1) Introductory concepts of mathematics - R. Cowen & E. M. Lungu, Bay Publ, (ISBN 999102577), 1999. 2) Thomas' Calculus, Eleventh Edition, M. D. Weir, J. Hass and F. R. Giordano, Pearson.	
	Resources/Equipment	
Lecturing, Tutoring and assessing space		